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Institutional Uncertainty and Taxpayer Compliance

By JAMES ALM, BETTY JACKSON, AND MICHAEL MCKEE*

The standard prescription for increased tax compliance has long been more frequent audits and more stringent penalties. Recently, however, another potential government strategy has begun to emerge from theoretical analysis of the compliance decision: taxpayer uncertainty about enforcement and tax policies. Such uncertainty may stem from several sources, such as imprecision in the tax code, lack of uniform training of government auditors, frequent changes in the tax code, and taxpayer ignorance of the factors that trigger an audit. In the face of such uncertainty, it is argued that a rational individual may respond by reporting more income than he or she would otherwise. In fact, there is some evidence that the Internal Revenue Service (IRS) believes this prediction and has deliberately maintained some uncertainty in its audit selection process in order to increase compliance.¹ There is also some evidence that taxpayers are aware of the uncertainty in IRS tax assessments. A survey of taxpayers by Yankelovich, Skelly, and White, Inc.

(1984) reports that 52 percent of taxpayers feel that IRS decisions are not consistent across taxpayers in the same circumstances.

The intuition here is straightforward. In making the compliance decision, an individual faces a portfolio-choice problem, with evasion as the risky, high-return asset and disclosure as the riskless asset. Suppose now that, say, the amount of unreported income that will be discovered by government audit becomes more uncertain. The greater risk on evasion income increases the cost of evasion while leaving unchanged the return to reported income, and an individual can decrease his or her exposure to this risk by complying more fully with the tax code. As shown by Alm (1988), Paul J. Beck and Woon-Oh Jung (1989), and Suzanne Scotchmer and Joel Slemrod (1989), an individual who maximizes the expected utility of the evasion gamble will often report more income.²

There are, however, several problems with the conclusion that taxpayer uncertainty increases compliance. First, it is largely based upon theoretical analysis, and at this point there is little other evidence to support this hypothesis. An exception is experimental work by Michael W. Spicer and J. Everett Thomas (1982) and Nehemiah Friedland (1982), who examine the impact of uncertainty about the probability of detection on compliance. However, they find conflicting evidence, and they do not investigate the effects of uncertainty about other fiscal parameters, such as the tax rate and the penalty rate. Beck et al. (1991) also use experiments to examine income uncertainty, and they find that such uncertainty increases reported income. However, like

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¹See *Roberts v. IRS* (1984), as cited in Jennifer F. Reinganum and Louis L. Wilde (1988).

²Reinganum and Wilde (1988) and Scotchmer (1989) also show that compliance may increase with greater income uncertainty in models that do not assume risk-averse taxpayers.

Spicer and Thomas (1982) and Friedland (1982), they do not examine other types of uncertainty, and their focus is on "taxpayer aggressiveness," or the willingness to report low income when income may be either high or low and when each has legal justification, not on taxpayer compliance.

Of perhaps more importance, the theoretical and experimental work ignores much accumulating evidence that tax compliance depends in part upon the use of the tax revenues.³ Ignoring government expenditures means that the individual compliance decision can be treated as independent across individuals, so that each individual need take only his or her own behavior into account when deciding to report income. In practice, however, an individual's choice may not be independent of the choices of others. To the extent that each individual recognizes that his or her taxes—and those of others—are used to provide public goods, then the presence of public goods makes behavior interdependent and implies that each individual must guess the behavior of other group members, even in large groups. In such an institutional setting, the impact of fiscal uncertainty need not be the same as when individuals receive nothing for their taxes.

This paper uses laboratory experiments to examine the effects of uncertainty on taxpayer compliance. The role of fiscal uncertainty is analyzed by comparing the compliance behavior of individuals when the key fiscal parameters (the tax rate, the probability of detection, and the penalty rate) are known with certainty against compliance when each of these parameters is made uncertain by randomly drawing the value of the parameter from a known distribution. The impact of government expenditures is also examined by introducing a public good in some sessions.

³Most of this work looks at voluntary contributions to public goods; see Robyn Dawes and Richard Thaler (1988) for a survey of the theoretical and experimental literature. Other work looks directly at the impact of government expenditures on compliance; for example, see Winfried Becker et al. (1987), Frank A. Cowell and James P. F. Gordon (1988), and Alm et al. (1992).

The results indicate that the impact of greater fiscal uncertainty depends upon the institutional setting in which the individual makes the compliance decision. When the decision is made independently of the use of the tax revenues, greater uncertainty always increases tax compliance, as predicted by much of the existing literature. However, if the fiscal institution specifies that individuals receive a benefit from government for their tax payments, then the introduction of uncertainty always leads to a fall in compliance. The effect of uncertainty therefore depends critically upon the way in which an individual's tax payments are linked with those of other group members in the governmental provision of goods and services. The stronger the link between the payment of taxes and the receipt of expenditures, the more compliance will suffer with greater uncertainty. It therefore appears that a government policy of allowing (or introducing) uncertainty in the voluntary-compliance system is a risky one that may have effects opposite of those expected.

I. Taxpayer Response to Uncertainty

The effects of institutional uncertainty upon taxpayer compliance can be easily demonstrated using the standard model of compliance. Suppose that an individual receives a fixed amount of income I and must choose how much to declare to the tax authorities. Declared income D is taxed at the rate t . Unreported income is not taxed; however, the individual may be audited with probability p , at which point a fine f is imposed on each dollar of unpaid taxes. If underreporting is detected, the individual's income I_C is

$$(1) \quad I_C = I - tD - ft(I - D)$$

while if underreporting is not detected income I_N is

$$(2) \quad I_N = I - tD.$$

The individual chooses D to maximize the expected utility $EU(I)$ of the evasion gam-

ble, or

$$(3) \quad EU(I) = pU(I_C) + (1-p)U(I_N)$$

where utility $U(I)$ is assumed to be a function only of income. This optimization generates the first-order condition

$$(4) \quad pU'(I_C)(f-1)t - (1-p)U'(I_N)t = 0$$

where a prime denotes a partial derivative.

Now let there be an "increase in risk" in one of the fiscal parameters in the individual's compliance decision (t , p , or f), as measured by a mean-preserving spread in the parameter.⁴ For example, suppose that the fine rate f becomes riskier. The impact on declared income of greater f risk depends upon the concavity or convexity in f of equation (4). If equation (4) is a convex (concave) function of f , then greater f risk will increase (decrease) D .

It is straightforward to show that the effects of f and t uncertainty are generally ambiguous.⁵ However, greater f risk will increase declared income if absolute risk aversion is nonincreasing, and greater t risk will increase D if relative risk aversion is nondecreasing and less than 1.⁶ Because expected utility is linear in the probabilities, greater p risk is predicted to have no effect on D .⁷

⁴Michael Rothschild and Joseph E. Stiglitz (1970) show that one random variable is riskier than a second one with the same expected value if the distribution of the first random variable can be obtained from that of the second by a mean-preserving spread.

⁵For example, tax-rate uncertainty has opposing effects on the incentive to declare income: tax-rate risk makes the tax payment on declared income uncertain but also makes the penalty on unreported income uncertain. Fine-rate uncertainty affects only the return to unreported income. Both t and f risk also have income effects that depend upon the individual's attitude toward risk.

⁶See also Alm (1988), Beck and Jung (1989), and Scotchmer and Slemrod (1989).

⁷Note, however, that there is growing evidence that individuals do not behave according to the axioms of the expected-utility hypothesis. In particular, there is much work that suggests that preferences are not linear in the probabilities. If the probability of detection enters the preference function in a nonlinear way, then greater p risk will affect the individual's choice of

These results are, however, altered by the introduction of government expenditures. Suppose that the government sums all tax payments, increases this sum by a multiple m to reflect the consumers' surplus from a public good, and distributes the resulting public good in equal shares s to all individuals. Denoting the reported incomes of all other individuals as G , the uncertain income of the individual is increased by $smt(D+G)$ in either state of the world.

The compliance decision of each individual is now affected by his or her beliefs about the contributions of others, and the impact of uncertainty now depends on the individual's beliefs about the responses of others via G , both to a change in his or her own compliance and to a change in institutional uncertainty. If the individual assumes that G is fixed and independent of these variables, then the impact of fiscal uncertainty is unchanged from the earlier analysis. It seems likely, however, that G is not fixed but instead depends upon f , t , p , and, perhaps, D . For example, Todd Sandler et al. (1987) show that increased uncertainty about the contributions of others to a public good may lower an individual's own contributions, and R. Mark Isaac, David Schmitz, and James M. Walker (1989) find that group contributions to a public good can increase if costs are reliably imposed on free riders. These results suggest that the tax payments of others will be affected by changes in t , f , and p . There is also some reason to expect that G will depend upon D , at least in small groups. Much theoretical and experimental research on the voluntary provision of public goods suggests that it is unrealistic to assume that an individual will always obey the Cournot-Nash conjecture, in which the individual believes that others will not react to a change in his or her behavior. In a single-shot game, Isaac, et al. (1985) find that subjects in the laboratory often contribute positive amounts to a public good even when there is no policing of free rid-

declared income. See, for example, Mark Machina (1987) for a discussion of attempts to generalize the expected-utility model.

ers. When the game is repeated, Michael Taylor (1986) finds that the potential for the cooperative outcome to emerge becomes greater, and Larry Samuelson (1987) shows that merely the expectation of continued play is often sufficient to induce cooperative behavior. For these reasons, it seems likely that G may also depend upon D .

The presence of a public good considerably complicates the analysis of greater uncertainty, making it impossible to determine a priori the impact of uncertainty upon compliance.⁸ In particular, the presence of

⁸For example, the presence of f risk introduces the terms G_D , G_{Df} , G_{Dff} , G_f , and G_{ff} into the expression that determines the convexity or concavity in f , where subscripts denote partial derivatives. The abandonment of the Cournot-Nash conjecture is reflected in the first three terms; the direct response of others' taxes to uncertainty is reflected in the last two terms. More precisely, the full expression for f risk becomes:

$$\begin{aligned} & \{pU'(I_C)smG_{Dff} + (1-p)U'(I_N)smG_{Dff} \\ & + 2pU''(I_C)[1 + smG_{Df}] \\ & \times [-t(I-D) + smtG_f] \\ & + 2(1-p)U''(I_N)[sm]^2tG_{Df}G_f \\ & + pU''(I_C)[f-1 + sm(1+G_D)]smtG_{ff} \\ & - (1-p)U''(I_N)[1-sm(1+G_D)]smtG_{ff} \\ & + pU'''[f-1 + sm(1+G_D)] \\ & \times [-t(I-D) + smtG_f]^2 \\ & - (1-p)U'''(I_N)[1-sm(1+G_D)] \\ & \times [smtG_f]^2\}. \end{aligned}$$

Even if others' contributions are not affected by the individual's choice of D , so that G_D , G_{Df} , and G_{Dff} all equal zero, the presence of G_f and G_{ff} makes it impossible to determine in general the concavity or the convexity of this expression. Similar conclusions hold for t risk. Note also that p risk may now affect the individual's compliance decision, even though expected utility is linear in the probabilities: as long as G depends upon p , then greater p risk introduces the terms G_D , G_{Dp} , G_{Dpp} , G_p , and G_{pp} into the relevant expression. Again, it is impossible to determine in general the sign of this expression.

the public good means that an individual must determine both his or her own response to greater uncertainty and the responses of all others. Under these circumstances, it is plausible that an individual may now react to greater uncertainty by lowering, not raising, his or her compliance. For example, as implied by Isaac, Schmidt, and Walker (1989), fine- or audit-rate uncertainty may reduce compliance because individuals will perceive the penalties for free riding to be less reliable. Similarly, as suggested by Sandler et al. (1987), tax-rate uncertainty may reduce compliance by making group contributions more uncertain: if the individual expects other to pay more, then he or she may be tempted to free ride, while if the individual expects others to pay less, then he or she may decide that paying one's own taxes is not worth the cost. Perhaps surprisingly, even if the individual expects all others' contributions to increase, his or her own compliance may still decline, since the individual's income from the public good will increase with greater contributions and compliance falls with greater income if there is decreasing absolute risk aversion.

In sum, the impact of fiscal uncertainty depends upon the institutional setting in which the compliance decision is made. When there is no public good, the individual need determine only his or her response to uncertainty. The addition of a public good means that the taxpayer must also determine how others will respond. The effects of uncertainty seem likely to differ in these two settings.

II. Experimental Design

The experimental design replicates the basic structure of the voluntary-income-tax system in the United States: subjects receive income, they pay taxes on income voluntarily reported, they face a probability of audit, and if they are detected cheating, they pay a penalty on taxes not reported. The effect of public-good provision on compliance is investigated by providing the subjects in some treatments with a public good whose size depends upon the tax payments of all indi-

viduals. In this institution there are three fiscal parameters that affect decisions on tax compliance: the tax rate, the probability of detection, and the penalty (or fine) rate.⁹

In each session the subjects are organized into a group of five. They are informed that the session will last for an unknown number of periods called rounds; in practice the actual number of rounds is predetermined at 25. Each subject is given an initial endowment of 10 tokens and is told that tokens accumulated during the experiment will be redeemed at an announced exchange rate for dollars at the end of the session. At the beginning of each round, each subject receives a random amount of income, drawn from the interval 2.00–3.00 in increments of 0.25. Only the individual knows his or her actual income. The individual is then asked to choose an amount of income to declare, and he or she pays taxes on the declared income.

An individual does not pay taxes on income not declared. However, each individual faces a chance that evasion will be detected and penalized. After all individuals have reported their incomes and paid their taxes in a round, up to one person is drawn at random for an audit. If a subject is audited, then the actual and declared incomes for the previous five rounds are compared; if the subject did not fully comply, any back taxes are collected and a fine equal to some multiple of unpaid taxes is imposed. The audit selection is performed using a bingo cage that contains one numbered ball for each subject plus additional balls that determine the actual probability of an audit. Balls are placed in and drawn from the cage in full view of the subjects. At the completion of each round, the subjects are shown their new balances.

⁹The instructions are available upon request. Unlike much previous experimental work on tax compliance, the instructions use “neutral” terminology; that is, all references to taxes, audits, declared income, and so on are replaced with more neutral words such as contributions, checks, disclosed income, and so on. The use of neutral instructions is standard practice in experimental methods. The discussion in the text uses tax terminology for ease of exposition.

In those treatments in which a public good is provided, it is introduced in a manner like that commonly used in the experimental literature on the voluntary provision of public goods.¹⁰ The taxes of all five individuals in the group are paid into a group fund, this fund is increased by a multiple of 2, and the resulting amount is then distributed in equal shares to the members of the group. Any amounts collected via audits are not added to the group fund. In those treatments without the public good, individuals receive nothing for their tax payments.

Several sessions are conducted, in which parameter values are chosen to correspond with real-world levels. The features of these sessions are summarized in Table 1. In the first session (S1 or the “base case”), all parameters are known with certainty. The probability of detection is 0.04, the penalty or fine rate is 2, and the tax rate is 0.30. Other sessions apply a mean-preserving spread to the variable of interest, with the subjects forced to make their compliance decision before knowing the realized value of the relevant variable. In all cases, the expected value of the random variable equals its certain value in the base-case session. Also, all treatments are run twice, once with a public good and once without.

In session 2 (S2 or “fine uncertainty”), the fine takes a value between 1.0 and 3.0 with equal probability.¹¹ In session 3 (S3, or “tax uncertainty”), the tax rate is a random variable drawn from a uniform distribution over the range 0.10 to 0.50.¹² In session 4 (S4, or “detection uncertainty”), the proba-

¹⁰See, for example, Isaac et al. (1985); Dawes and Thaler (1988) discuss much of this literature.

¹¹Twenty balls numbered 10–30 are placed in a bingo cage. After the subjects declare their incomes and a subject is selected for audit, one ball is drawn from the cage. The fine rate equals the number on the ball divided by 10 (e.g., ball number 10 yields a fine rate of 1.0, etc.). The fine rate is announced at the end of the round.

¹²Forty balls numbered 10–50 are placed in a bingo cage. After the subjects declare their incomes, one ball is drawn from the cage. The tax rate equals the number on the ball (e.g., ball number 10 yields a tax rate of 10 percent, etc.). As in S2, the tax rate is announced at the end of the round.

TABLE 1—EXPERIMENTAL DESIGN

Session	Fine rate	Tax rate	Probability
S1: base case	2	0.30	0.04
S2: fine uncertainty	1.0–3.0 (mean = 2.0)	0.30	0.04
S3: tax uncertainty	2	0.10–0.50 (mean = 0.30)	0.04
S4: detection uncertainty	2	0.30	0.02 or 0.06 (mean = 0.04)

Note: All sessions are run twice—once with the public good and once without. When there is a public good, the multiplier on the group fund equals 2.

bility of detection is a random variable that equals 0.02 or 0.06 with equal probability.¹³

Subjects are volunteers from Principles of Economics classes and have no prior experience with the experimental setting (including any form of public-goods experiments). Each treatment is replicated three times with different subjects, all run concurrently in the laboratory. Each session lasts approximately one hour, and the average payoff per subject varies from \$15 to \$25.

The experiments are conducted in a computerized laboratory equipped with 15 subject terminals and a monitor terminal. All subject entries are made on the terminals, all information produced during the experiment is presented on the subjects' terminals, and all calculations are performed by the computer.

¹³Two bingo cages are used to determine the probability of audit. In one bingo cage there are 17 balls numbered 1–17, so that the probability of detection is 0.06; in the other cage there are 50 balls numbered 1–50, which gives a probability of 0.02. (Recall that each session is replicated with three different groups of five subjects. These replications are run concurrently, so that the 15 subjects are in the laboratory at the same time.) A six-sided die is used to determine which cage is selected (e.g., if an odd number is rolled, then the first cage is used to select a numbered ball). If a ball numbered 1–15 is drawn, then the subject with that identification number is audited; if a ball numbered 16 or greater is drawn, then no one is audited. Note again that an individual who maximizes expected utility would not be affected by the introduction of detection uncertainty, since expected utility is linear in the probabilities. However, there is much evidence that individuals do not always behave according to the axioms of expected utility.

Note that this experimental design differs from that of some earlier experimental work. The values of the various policy parameters used here are chosen to approximate real-world values. Further, the results are verified with repeated experiments; each session is replicated three times, and the reported results are averages over these three replications. Finally, the instructions given to subjects use neutral terminology, not tax terminology, in order to avoid context or framing effects that may bias subject choices unpredictably.

III. Results

The results are summarized in Table 2. All statistical comparisons are made using difference-of-means tests on the average compliance rates for each treatment. Statistical analysis of the difference in means requires that the data on average compliance rates be normally and independently distributed. The *individual* compliance rates over the 25 rounds of the experiment may not be independently distributed, especially in those sessions in which a public good is present. However, independence can be achieved by taking *group* averages of the compliance rates in each round, across the three groups in each session. Specifically, the average compliance rate for a given session and round is calculated in two steps: first, by averaging for each round the compliance rates across all five subjects in a group; and second, by taking the average of the three group averages. This procedure

TABLE 2—AVERAGE COMPLIANCE RATES AND SUMMARY STATISTICS

Session	Mean	Standard deviation
No public good:		
S1: base case	0.262	0.067
S2: fine uncertainty	0.374	0.082
S3: tax uncertainty	0.370	0.067
S4: detection uncertainty	0.481	0.087
Public good:		
S1: base case	0.557	0.072
S2: fine uncertainty	0.501	0.082
S3: tax uncertainty	0.398	0.067
S4: detection uncertainty	0.519	0.074

generates 25 observations for each session, or one for each round of the session. These observations are independent because they are averages across three unrelated groups of subjects. Normality of these observations is tested using the Bowman-Shelton statistic (Paul Newbold, 1988). Means and standard deviations are computed from these 25 observations, and it is these values that are reported and used in the hypothesis tests.¹⁴ Although the average compliance rate varies across the different sessions, there is substantial compliance in each session that generally does not decay as the experiment proceeds.

Consider first the introduction of uncertainty in the sessions with no public good. Uncertainty always leads to increased compliance, as predicted by much of the theoretical literature. The average compliance rate is 0.262 in the base-case session (S1), while it rises to 0.374 in the fine-uncertainty session (S2), a difference that is statistically significant ($t = 5.21$). Uncertainty about the tax rate also increases compliance in the absence of the public good. Average compliance rises from 0.262 in the base case to 0.370 in the tax-uncertainty case (S3), an

increase that is statistically significant ($t = 5.65$). Finally, uncertainty about the audit rate increases compliance from 0.262 in the base case to 0.481 in the detection-uncertainty session (S4), a difference that is both large and statistically significant ($t = 9.88$).¹⁵

Consider now those sessions in which a public good is provided with the tax revenues. Note that average compliance in any session is always higher in the presence of the public good, so that individuals pay more taxes when they receive something for their taxes.¹⁶ However, unlike the sessions with no public good, the introduction of fiscal uncertainty in the presence of a public good always lowers the average compliance rate relative to the base case. Compliance is higher in the public-good base case (0.557) than in the fine-uncertainty session (0.501), a difference that is statistically significant ($t = 1.80$). Tax-rate uncertainty also significantly lowers compliance from 0.557 in the base case to 0.398 in the tax-uncertainty session ($t = 4.26$). Compliance falls when the probability of detection is made uncertain; the difference between the base case and the detection-uncertainty session (0.519) is not large but is still statistically significant ($t = 1.86$).

It is possible that the above results for the public-good sessions arise because of the small group size in the experiments. In small groups, group contributions are likely to respond to changes in individual behavior.¹⁷ In large groups, the impact of a single individual's contributions on the level of public-good provision is negligible, and the

¹⁴ Individual data are available upon request. All hypothesis tests are also conducted using the Mann-Whitney nonparametric test (W. J. Conover, 1980), which does not require normality of the data. The Mann-Whitney tests yield the same results as the difference-in-means tests and so are not reported.

¹⁵ There is also much evidence of all-or-none behavior, in which an individual either fully complies or fully cheats, as would result from any linear (or risk-neutral) payoff function. Overall, roughly two-thirds of the individual observations are all-or-none, although this varies with the session.

¹⁶ Becker et al. (1987) and Cowell and Gordon (1988) also find that compliance is affected by government expenditures. Other experimental work shows that compliance increases as the group-fund multiplier increases (Alm et al., 1992).

¹⁷ In fact, there is a significant positive correlation between an individual's compliance in a round and the amount of the public good received in the previous round.

effect of uncertainty in the presence of a public good may well approach its impact in the absence of government expenditures as group size increases. However, Isaac, Walker, and Arlington W. Williams (1989) show that group size does not have a significant effect on group contributions; that is, voluntary contributions in small groups are comparable to contributions in large groups. More importantly, it is the mere presence of the public good that is likely to affect individual compliance, and this effect is independent of the group size. Even if an individual's choice of D has a negligible effect on group behavior, group contributions are almost certain to be affected by uncertainty about f , t , or p , and an individual must consider these effects on group behavior when making his or her individual choice.¹⁸ Because the individual must guess the group response, behavior becomes interdependent.

Note that each subject might be expected to gain information about the group response to greater uncertainty as the experiment proceeds. If subjects become more certain about group contributions over time, then the impact of this source of uncertainty upon individual compliance could diminish, and the average compliance rates in the various public-good sessions could converge. However, such convergence does not in fact occur. For example, if comparisons are made using the average group compliance rates from only the last five rounds of each session, the results are not altered.¹⁹

The difference in results is striking but plausible. In the absence of a public good, an individual makes his or her compliance decision independently of the choices of others. However, in the presence of a public good whose magnitude depends upon the

contributions of all taxpayers, these decisions become interdependent, even in large groups; that is, an individual must now determine his or her own response to uncertainty and must also guess the responses of other group members. For example, individuals may well contribute less to a public good when there is uncertainty about group contributions, which arises here from tax-rate uncertainty.²⁰ Uncertainty about enforcement efforts (the probability of detection and the fine rate) may also lower compliance. If individuals believe that either fine or detection uncertainty makes enforcement against noncompliers less reliable, then they may well pay less in taxes.

IV. Conclusions

The experimental results provide mixed support for the hypothesis that greater uncertainty increases tax compliance. Introducing uncertainty in the available fiscal parameters affects compliance in opposite ways, depending upon the presence or absence of a public good. When individuals receive nothing for their tax payments, uncertainty increases compliance, a result consistent with much of the literature and the apparent views of the IRS. However, when individuals perceive that they receive a public good in exchange for their taxes, uncertainty always lowers compliance. In sum, a policy, intentional or not, of increasing the level of fiscal uncertainty is a risky tool for generating increased compliance. Individuals may well respond by reducing their compliance. If that occurs, then not only are individuals made worse off by the uncertainty, but the government may also lose tax revenues.

¹⁸More formally, as discussed in footnote 8, G is likely to depend upon, say, the fine rate f via the terms G_f and G_{ff} , even if G does not depend upon D via the terms G_D , G_{Df} , and G_{Dff} .

¹⁹The average compliance rates for the last five rounds of public-good sessions 1–4 are 0.606, 0.452, 0.338, and 0.519, respectively. Alternative cutoff points (e.g., the last ten rounds) give similar results.

²⁰Additional support for this result comes from another session in which there was uncertainty about the group-fund multiplier. The multiplier was drawn in each round from a uniform distribution with mean 2.0 and support 1.0–3.0. Average compliance declined significantly from 0.557 in the public-good base-case session to 0.485.

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